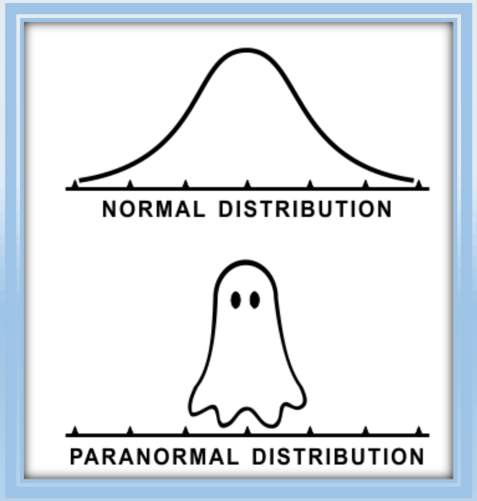
Mathematics II (part 1)

SAMPLES & SAMPLING DISTRIBUTION (part 2)



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SUMMARY of studies

Descriptive research or sample survey

- You randomly sample and collect data through surveys (or similar means).

Observational Studies

- A researcher uses observed information to learn about a group of individuals.
- The individuals being observed are specifically not interfered with aside from the measuring of their responses
- Such study doesn't establish causality because there could be confounding or lurking variables but can help establish **correlation**.

Randomized Controlled Experiment

- A researcher deliberately imposes different treatments on different groups and then compares the groups to determine if there is any difference in a specific variable of interest (called a response).
- Researchers can establish **causality** through this.

What are we going to learn?

Sampling methods	• Probability and Nonprobability sampling types
Inferring from samples	• Constructing a sampling distribution
Central limit theorem	• What does it mean? • Example of use
Smaller Samples	• T-distributions

SAMPLING METHODS

Probability and Nonprobability sampling

Sampling methods

Introduction

- Regardless of which type of study you will conduct; you need to collect data.
- Since access to all of a population is usually not possible, we collect samples.
- We need sample data to have variability, and data quality is as important as size.

What is the best way to sample?

Sampling methods

Examples

We would like to know if students are satisfied with SIT as an institution.

Q The president would like to know if students are satisfied with SIT as an institution. Which of the following will allow the president to make valid conclusions about the students' opinions?

- The president asks every student
- The president will write a random number generator program to make a list for asking 1000 students
- The president picks 100 students randomly from a list
- The president will ask students from computing
- The president will ask students closest to his/her office
- The president asks for people to volunteer to answer his question
- The president will ask students he/she knows

Sampling methods

Examples (Answers)

The president would like to know if students are satisfied with SIT as an institution. Which of the following will allow the president to make valid conclusions about the students' opinions?

A

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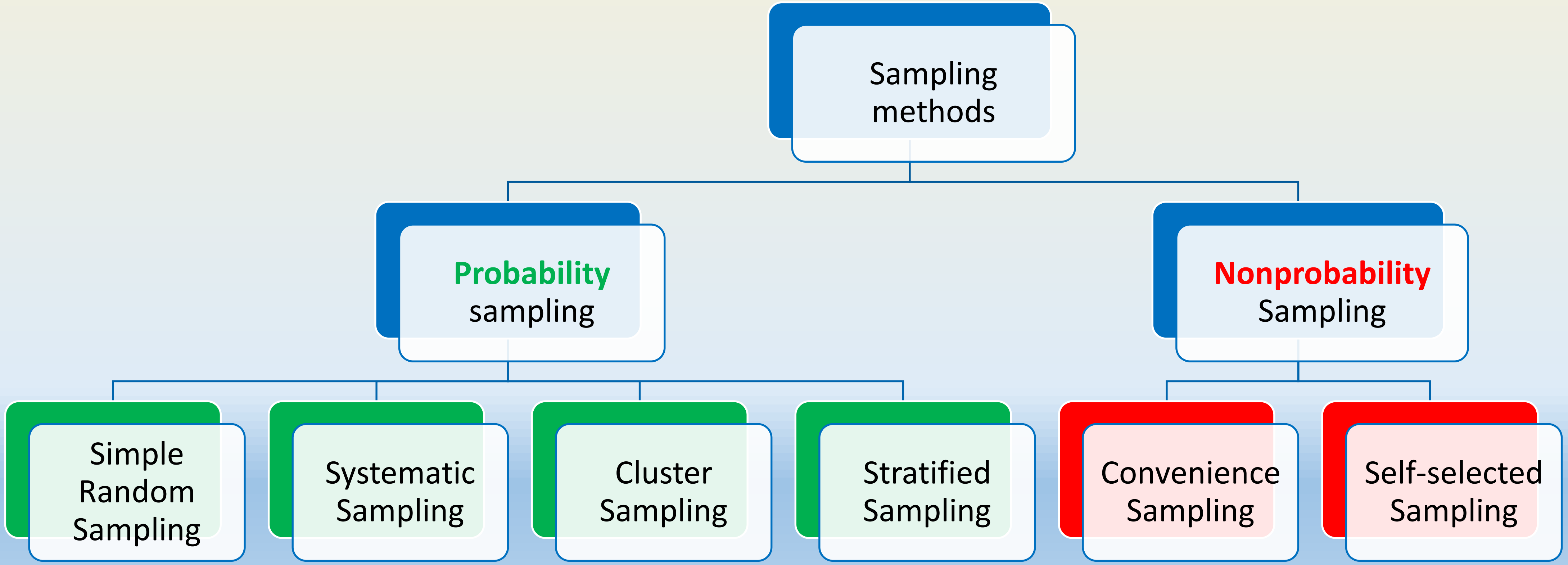
Sampling methods

Methods: Probability vs Nonprobability

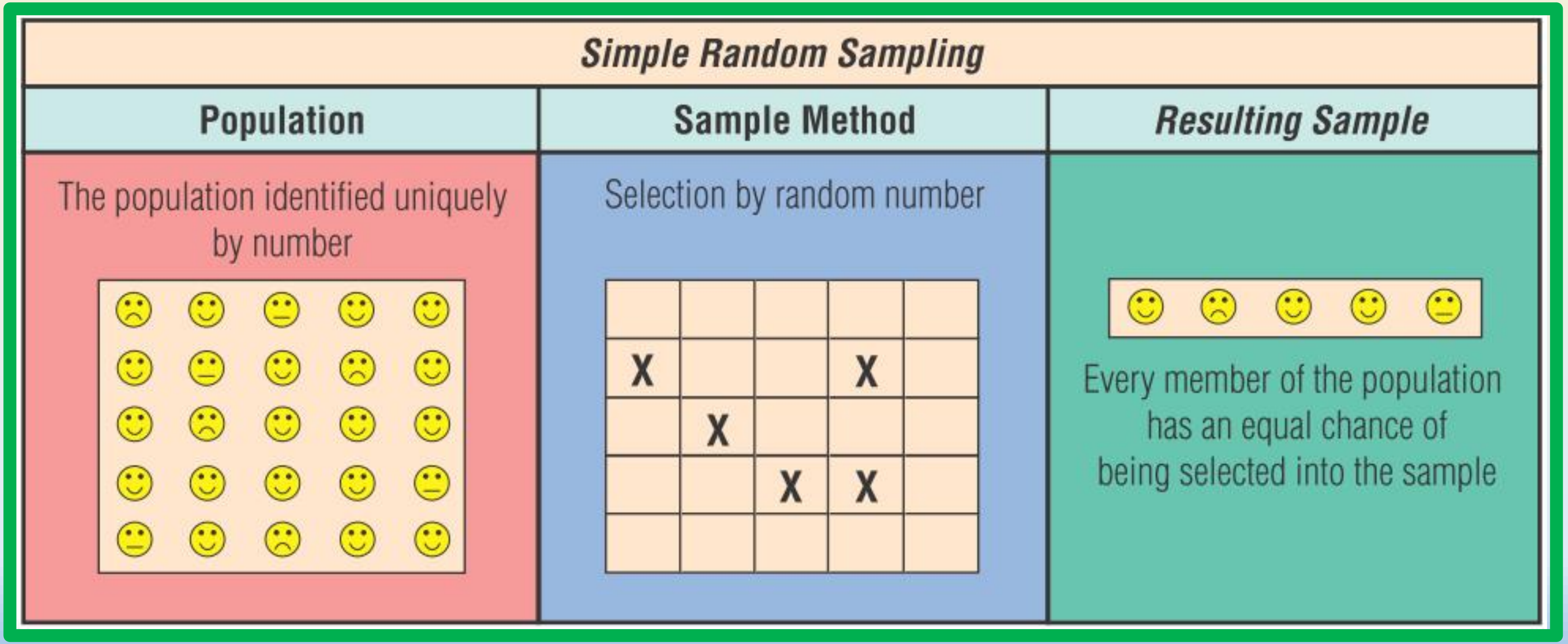
- How to collect reasonable samples?
- Some common types of sampling methods used in practice to collect a reasonable sample.



Sampling methods

Probability: Simple random sampling

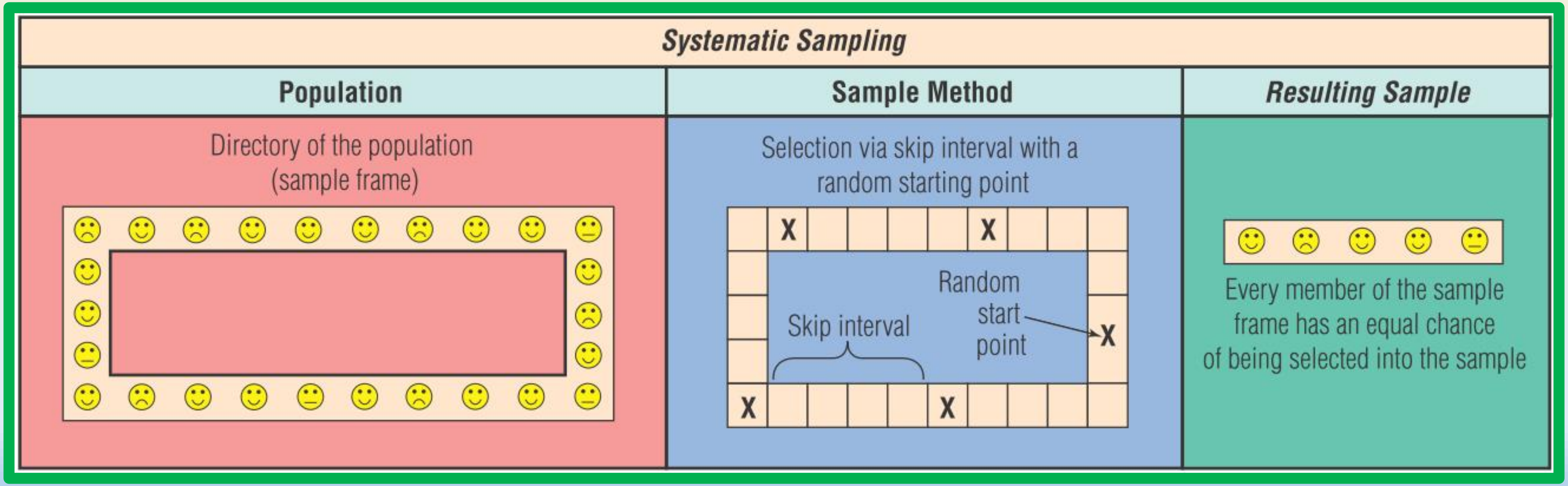
In **simple random sampling**, researcher uses **random numbers** to ensure every member of the population has an **equal chance** of being selected into the sample.



Sampling methods

Probability: Systematic sampling

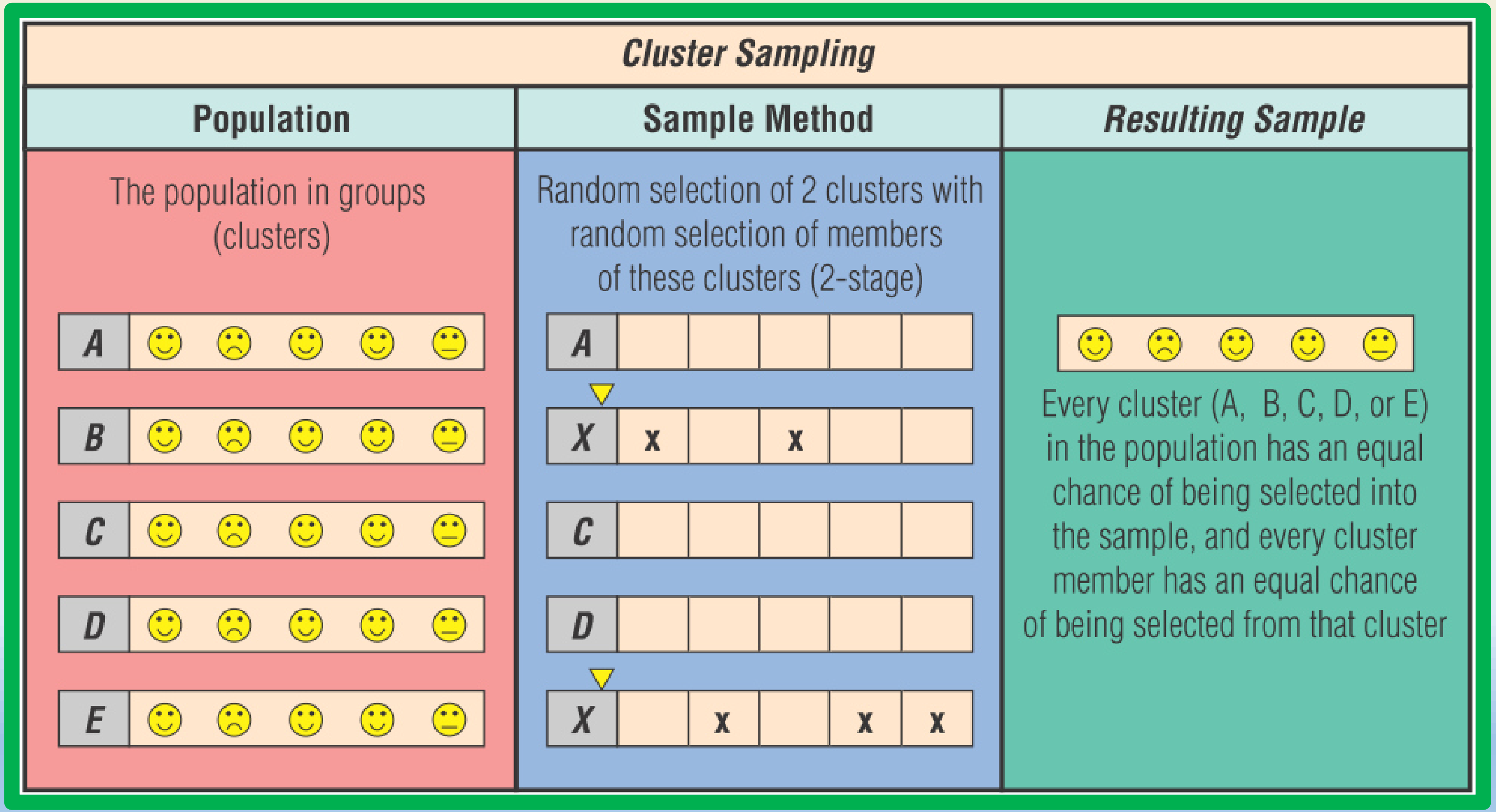
In **systematic sampling**, members are placed in a certain order. Researcher selects a **random starting point** for the first sample member, then **skip a constant interval** to select other members.



Sampling methods

Probability: Cluster sampling

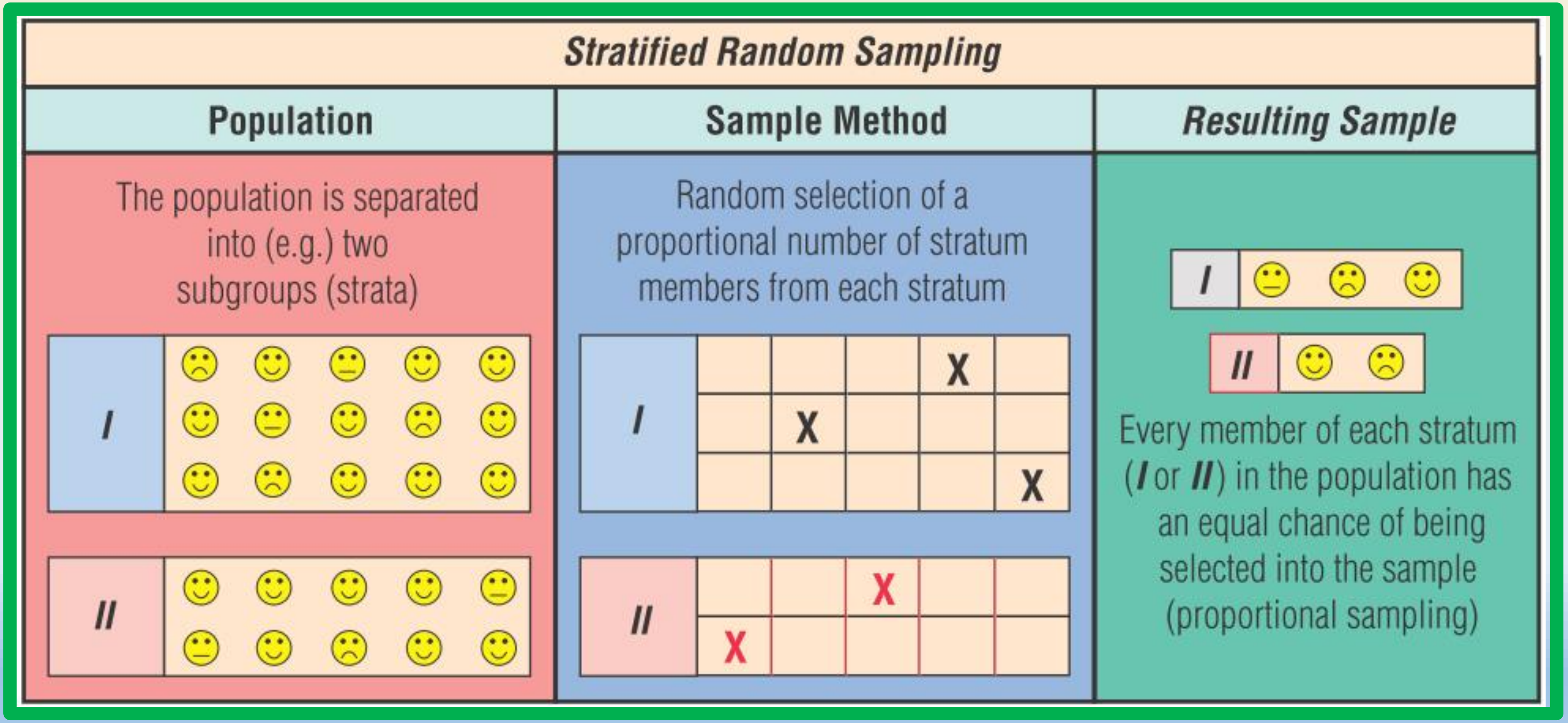
In **cluster sampling**, population is divided into **clusters** which are similar to other **clusters**. Then samples are **randomly selected** from one or more **randomly selected clusters**.



Sampling methods

Probability: Stratified random sampling

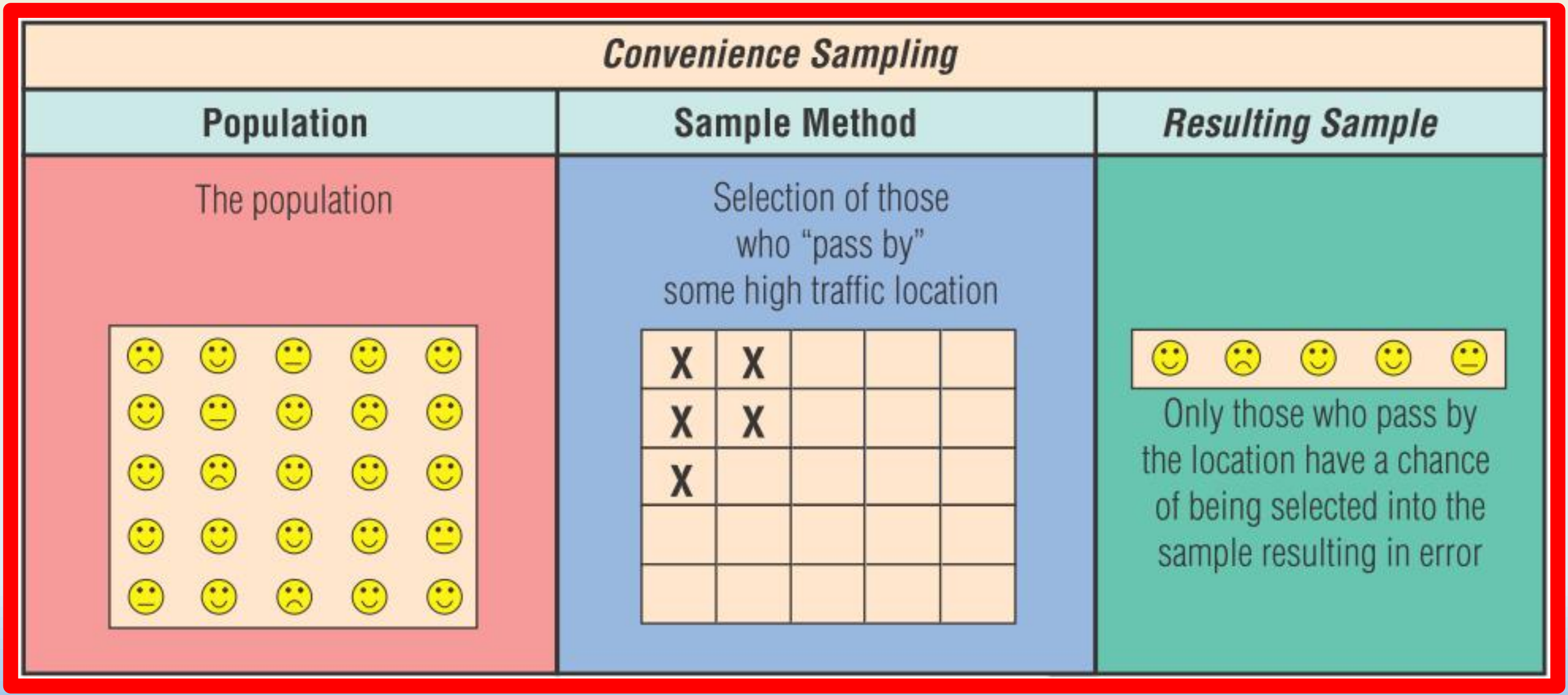
In **stratified random sampling**, population is **divided into different strata** (e.g., Income group, etc). A **simple random sampling** is then taken from **each stratum.**



Sampling methods

Nonprobability: Convenience sampling (haphazard)

In contrast, **nonprobability sampling methods** don't ensure every member of the population has an equal chance of being selected; **e.g. here convenience sampling**, etc. They should be avoided.



Sampling methods

Nonprobability: Self-selected e.g., Voluntarily response

- For example, relying on self-selected voluntarily response are sure to be **biased in favor of those with strong opinions**, making the result meaningless.
- A survey conducted by an author for her book “*Women and Love*”.
- She was reported to have sent questionnaires to 100,000 women of which about
- 4.5% responded.

Survey Finds Most Women Unhappy in Their Choice of Husbands

A popular women's magazine, in a survey of its subscribers, found that over 90% of them are unhappy in their choice of whom they married. Copies of the survey were mailed to the magazine's 100,000 subscribers. Surveys were returned by 5000 readers. Of those responding, 4520, or slightly over 90%, answered no to the question: “If you had it to do over again, would you marry the same man?” To keep the survey simple so that people would return it, only two other questions were asked. The second question was, “Do you think being married is better than being single?” Despite their unhappiness with their choice of spouse, 70% answered yes to this. The final question, “Do you think you will outlive your husband?” received a yes answer from 80% of the respondents. Because women generally live longer than men, and tend to marry men somewhat older than themselves, this response was not surprising. The magazine editors were at a loss to explain the huge proportion of women who would choose differently. The editor could only speculate: “I guess finding Mr. Right is much harder than anyone realized.”

Sampling methods

Match previous examples with Probability sampling method

1. A teachers puts students' names in a hat and chooses without looking to get a sample of students.
   - (Simple Random Sampling)
2. A student council surveys 100 students by getting random samples of 25 freshmen, 25 sophomores, 25 juniors, and 25 seniors.
   - (Stratified Sampling)
3. A principal orders t-shirts and wants to check some of them to make sure they were printed properly. She randomly selects 2 of the 10 boxes of shirts and checks every shirt in those 2 boxes.
   - (Cluster sampling)

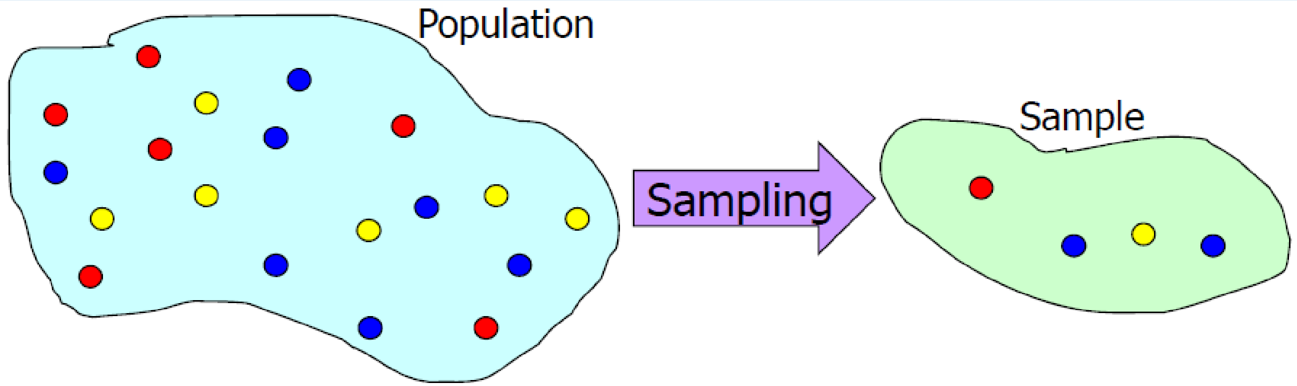
INFERRING FROM SAMPLES

HOW TO ACCURATELY INFER?

Inferring from samples

Recap of sampling

Based on collected data, calculate sample statistic to infer population parameter to answer our original curiosity question.



Parameter – a summary number describing a population, e.g.:

- mean μ
- variance σ^2
- standard deviation σ
- proportion p

Statistic – a summary number describing a sample, e.g.:

- mean $\overline{x}$
- variance s^2
- standard deviation s
- proportion $\hat{p}$

Note: Do not be confused with Statistics as a branch of mathematics and statistic to mean a summary number of a sample.

5

Probability distribution example

Example:

Consider a population of four numbers: 1, 3, 5 and 7

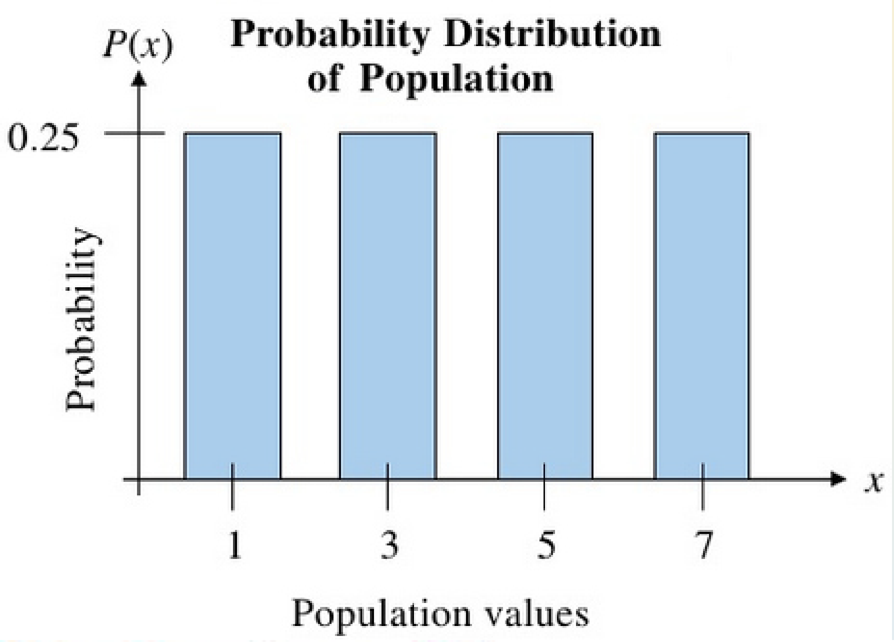
What is the expected value?

$E(X)$ = Mean = 4

What is the variance?

$Var(X)$ = Variance = 5

What if I pick a sample of size 2 or 3 from this population and calculate the mean for the sample, would it be the same?



Discrete random variable:

- mean $\mu = \sum x_i P(X = x_i)$
- variance $\sigma^2 = \sum (x_i - \mu)^2 P(X = x_i)$
- standard deviation $\sigma = \sqrt{\sigma^2}$

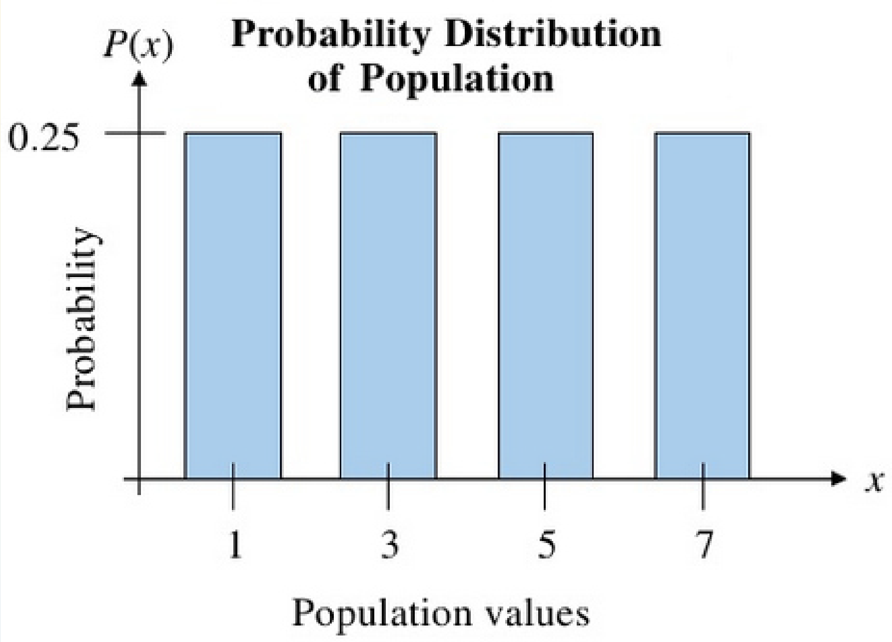
Inferring from samples

Probability distribution example

Example: Consider a population of four numbers: 1, 3, 5 and 7

E(X) = Mean = 4

Var(X) = Variance =5



Naturally, we would expect the calculated **sample statistics** to **vary from sample to sample** if different sets of sample data were to be collected.

Probability distribution example (is this true?)

Example: Consider the same population of four numbers: 1, 3, 5 and 7. If a random sample size **n = 2** is selected with replacement, find sample means

Sample	$\bar{x}$	Sample	$\bar{x}$
1, 1	1	5, 1	3
1, 3	2	5, 3	4
1, 5	3	5, 5	5
1, 7	4	5, 7	6
3, 1	2	7, 1	4
3, 3	3	7, 3	5
3, 5	4	7, 5	6
3, 7	5	7, 7	7

Constructing a sampling distribution

Example: Consider the same population of four numbers: 1, 3, 5 and 7. If a random sample size **n = 2** is selected with replacement, find sample means

From probability theory, we can let a random variable be the sample statistic, e.g. sample mean. Then, we can construct a distribution of sample statistic called sampling distribution.

Sample	$\bar{x}$	Sample	$\bar{x}$
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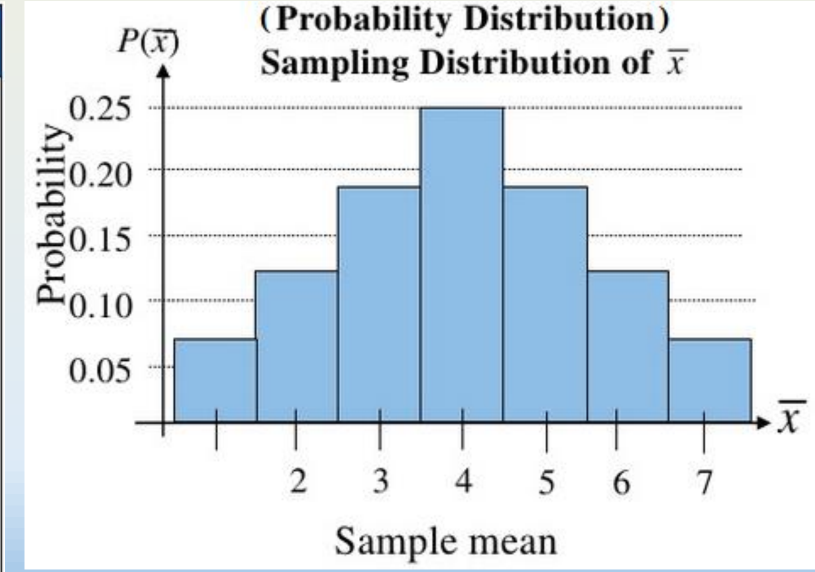
1, 7	4	5, 7	6
3, 1	2	7, 1	4
3, 3	3	7, 3	5
3, 5	4	7, 5	6
3, 7	5	7, 7	7

$\bar{x}$	f	Probability
1	1	0.0625
2	2	0.1250
3	3	0.1875
4	4	0.2500
5	3	0.1875
6	2	0.1250
7	1	0.0625



Constructing a sampling distribution

Since **sampling distribution** is a probability distribution, it therefore also has expected value and variance.

What is the expected value (mean)?

$\bar{x}$	f	Probability
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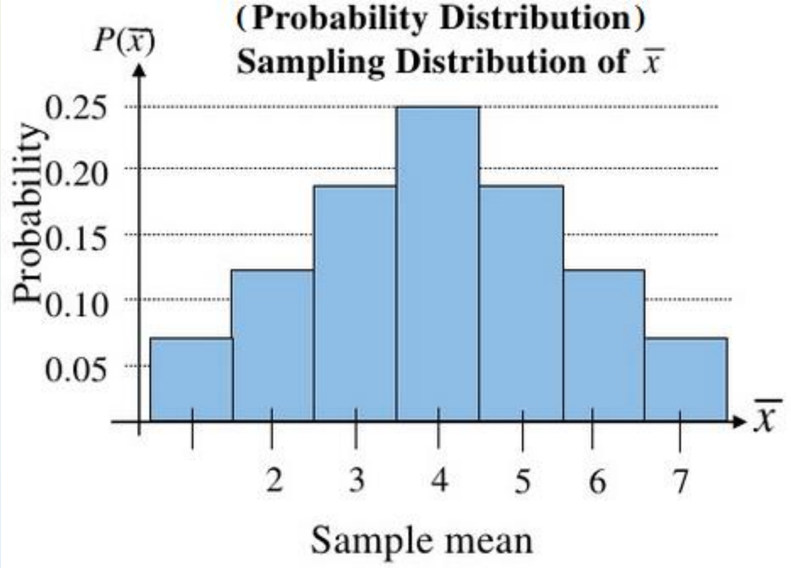

Inferring from samples

Constructing a sampling distribution

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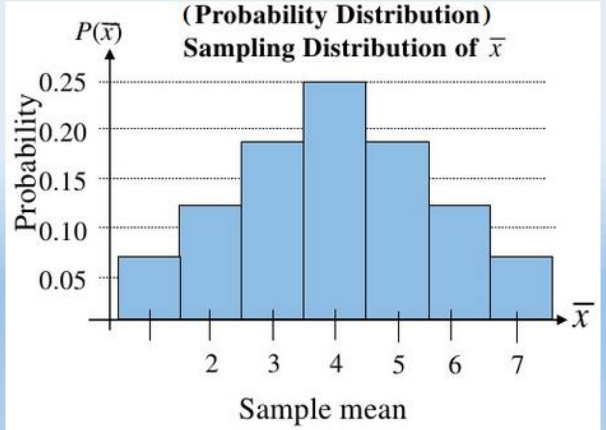


Solution:

Mean of the sampling distribution of $\bar{X}$:

$$E(\bar{X}) = \mu_{\bar{X}} = \sum \bar{x} P(\bar{X} = \bar{x})$$

$$= 1(0.0625) + 2(0.125) + \ldots + 7(0.0625) = 4$$

Variance of the sampling distribution of $\bar{X}$:

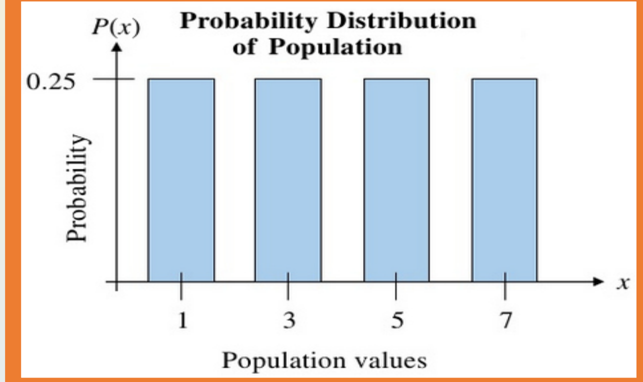
$$E(\bar{X}^2) = \sum \bar{x}^2 P(\bar{X} = \bar{x})$$

$$= 1^2(0.0625) + 2^2(0.125) + \ldots + 7^2(0.0625) = 18.5$$

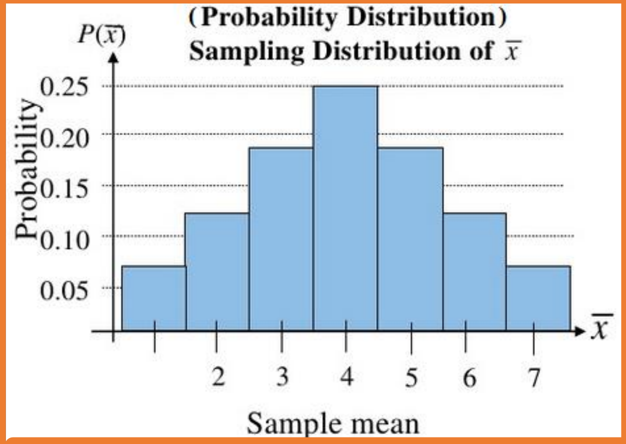
$$Var(\bar{X}) = \sigma_{\bar{X}}^2 = E(\bar{X}^2) - \mu_{\bar{X}}^2$$

$$= 18.5 - 4^2 = 2.5$$

Anything different about the distributions' parameters?



$E(X) = 4$
$Var(X) = 5$



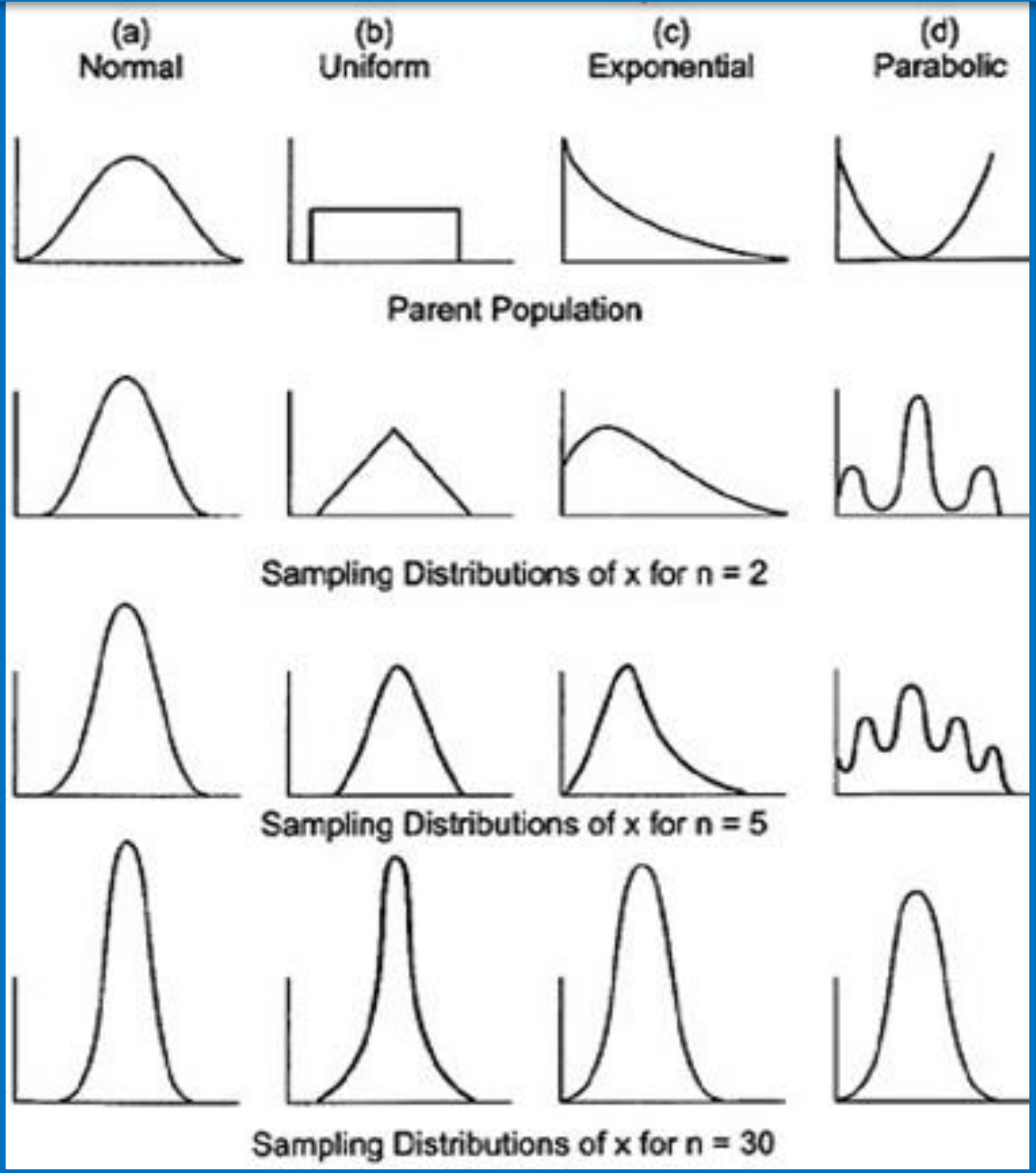
$E(\bar{X}) = 4$
$Var(\bar{X}) = 2.5$

CENTRAL LIMIT THEOREM

What is it?

What does it mean?

What does this figure tell us?

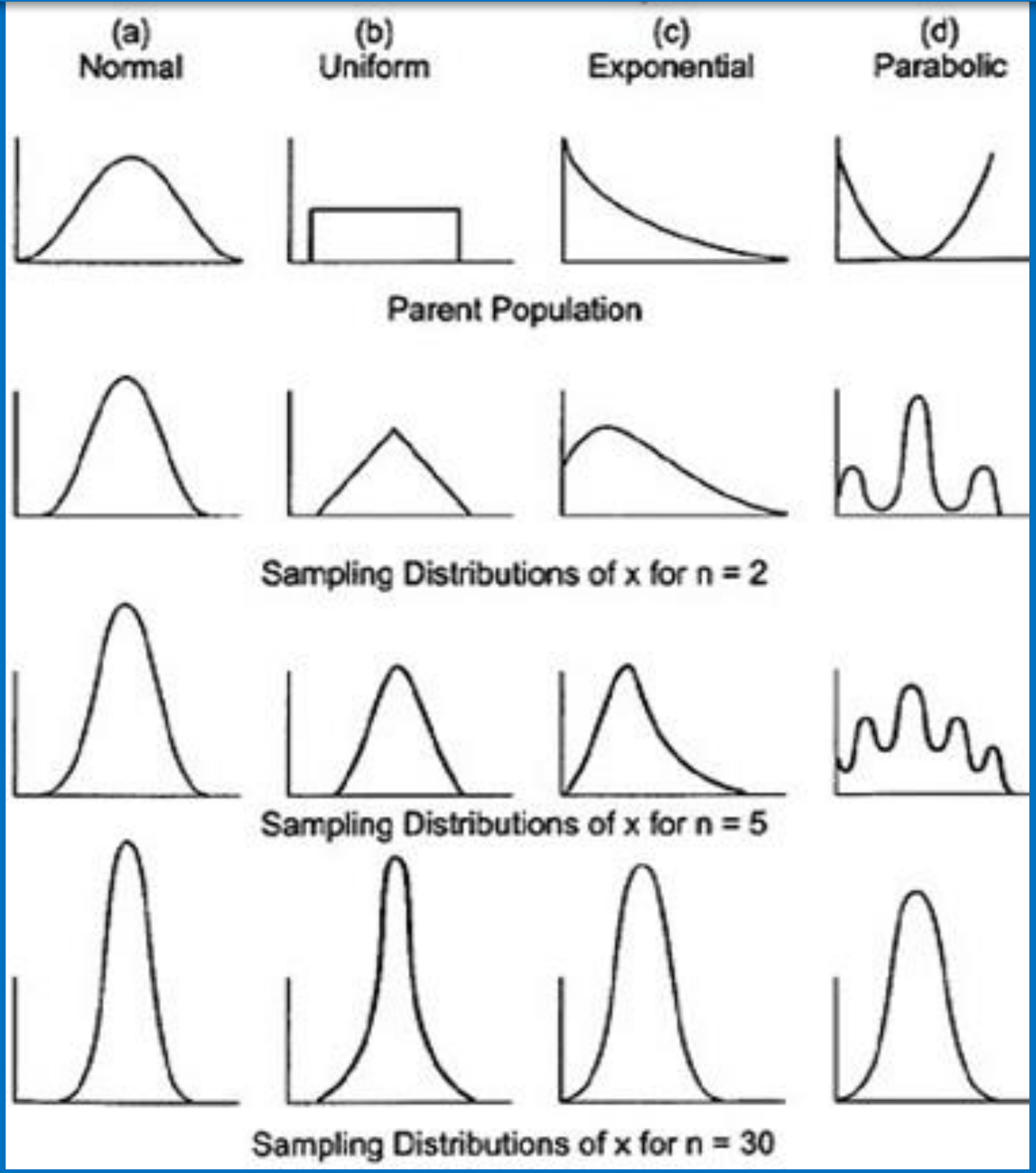


Q

Central limit theorem

What does it mean?

What does this figure tell us?



With increasing sample size, the distribution of sample means approach the normal distribution **regardless of the distribution of the parent population**.

Central limit theorem

Definition

"It has been observed that if **sample size, n, is sufficiently large**, the **sampling distribution** of sample mean is approximately normally distributed."

If a random sample of **sufficiently large size *n* (≥30)** is drawn from this population, the **sampling distribution** of the sample mean will have an approximately normal distribution with mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma/\sqrt{n}}$

Stdev of sample Distribution:

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

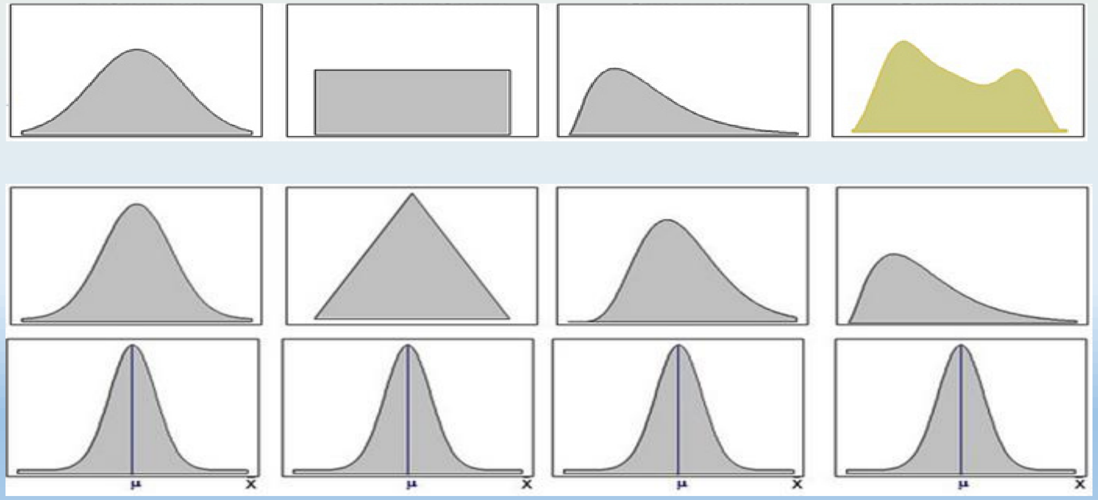
Central limit theorem

Approximating with central limit theorem

"It has been observed that if **sample size, n, is sufficiently large**, the **sampling distribution** of sample mean is approximately normally distributed."

If a random sample of **sufficiently large size *n* (≥30)** is drawn from this population, the **sampling distribution** of the sample mean will have an approximately normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$

- Original population distribution:
- **Sampling distribution** of sample mean:
  - **n=2**
  - **n>=30**



Central limit theorem

Example

Example: A cola filling machine is supposed to fill cans with 12oz of cola. Suppose that the fills are known to have a mean of 12.1oz and a standard deviation of 0.2oz. What is the probability that the sample mean for a 36-pack of cola is less than 12oz?

Q

Central limit theorem

Example

Example: A cola filling machine is supposed to fill cans with 12oz of cola. Suppose that the fills are known to have a mean of 12.1oz and a standard deviation of 0.2oz. What is the probability that the sample mean for a 36-pack of cola is less than 12oz?

Mean same for sample

Solution: Since n=36 is sufficiently large, CLT tells us that the sampling distribution of $\bar{X} \sim \boxed{N\left(12.1, \frac{0.2^2}{36}\right)}$.

$s^2 = \sigma^2/n$

We use *N* for a random pick from this probability

Now, we convert with Z for specific sample

$$Z = \frac{X - \mu}{\boxed{S}}$$

Central limit theorem

Example

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to standard normal

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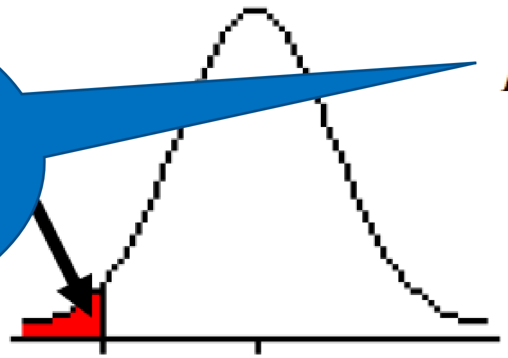


$$P(\bar{X} < 12) = P\left(Z < \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}\right) = P\left(Z < \frac{12-12.1}{0.2/\sqrt{36}}\right)$$

$$= P(Z < -3)$$

$$= 1 - P(Z < 3)$$

$$= 1 - 0.9987 = 0.0013$$

Now, we convert with Z for specific sample

$$Z = \frac{X-\mu}{\boxed{S}}$$

SMALLER SAMPLES

T-DISTRIBUTIONS

Smaller samples

What to do with smaller samples?

In practice, we rarely know the population variance required for CLT.

So can we use? $Z = \dfrac{\bar{x} - \mu}{s}$

✓Yes you can, **but the estimations will not be very good for small samples.**

Fortunately, there is another model for sampling distribution:

Student's *t*-distribution.

- Using this model, we can get good estimations with **degrees of freedom n-1**

$$Z < \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}} \rightarrow t = \dfrac{\bar{x} - \mu}{s / \sqrt{n}}$$

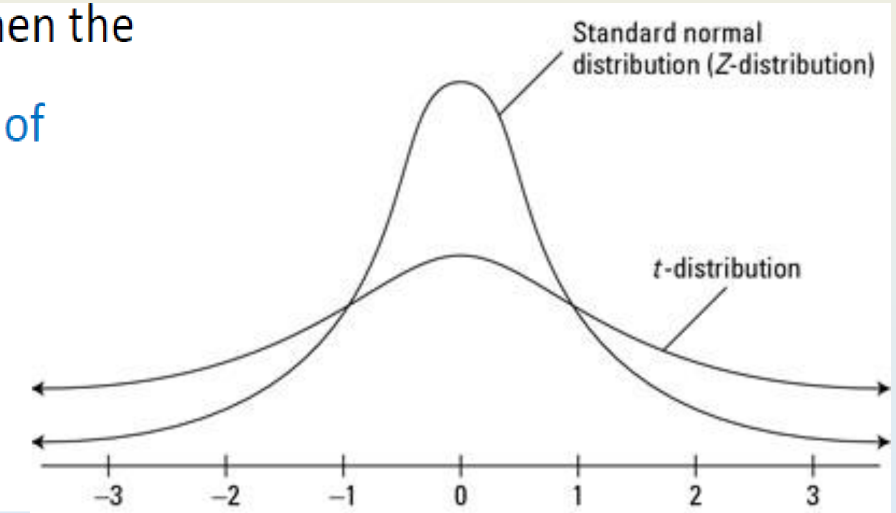
Smaller samples

What to do with smaller samples?

In probability and statistics, **Student's t-distribution** (or simply the t-distribution) is any member of a family of continuous probability distributions that arises when estimating the mean of a normally distributed population in situations where the sample size is small and population standard deviation is unknown. It was developed by William Sealy Gosset under the pseudonym Student (Wikipedia).

If a random sample of size n is drawn from this population, then the sample statistic $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$ has a t-distribution with degrees of freedom df = $n - 1$.

$$\boxed{Z < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}} \Rightarrow \boxed{t = \frac{\bar{x} - \mu}{s/\sqrt{n}}}$$



Smaller samples

Degrees of freedom

== *Number of **variables that can vary** in your sample.*

Imagine having **sample mean = 2** and the **size of sample is n = 3**

By default, we say that **degree of freedom** is **n-1 =2**

But, Why?

If I vary the **first variable** in sample and the **second variable** in sample, can I vary the **third one** and still get **mean 2?**

Smaller samples

Degrees of freedom

Imagine a sample **{a,b,c}** has a **mean 2**.

Many possible values can fit in this sample: **{1,2,3}** or **{2,2,2}**, etc.

But if we set **a=1** and **b=1**. What are the possibilities of **c**?

The only possible answer is **4**

This is why, when we have a sample of size **n variables**, our **degrees of freedom** are always **n-1**

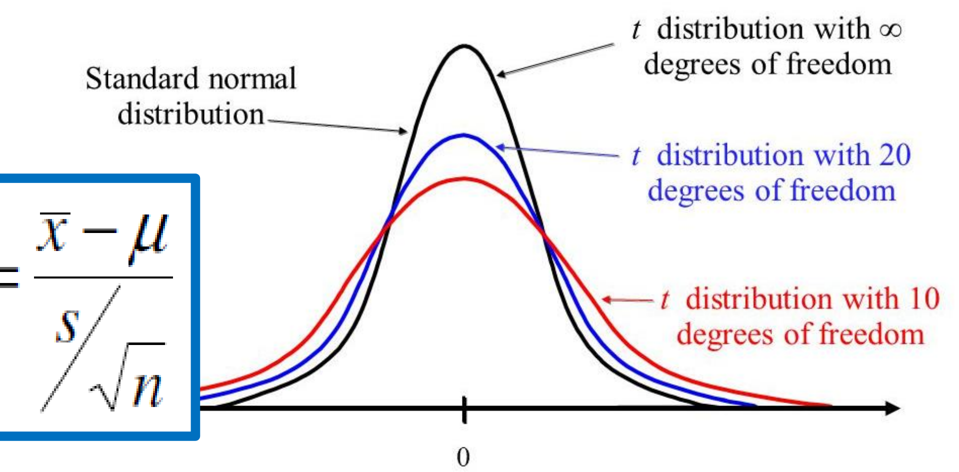
Smaller samples

T-tables

- Each Row is a different distribution
- With different **degree of freedom**

t Distribution

The t-distribution is used when n is **small** and σ is **unknown**.



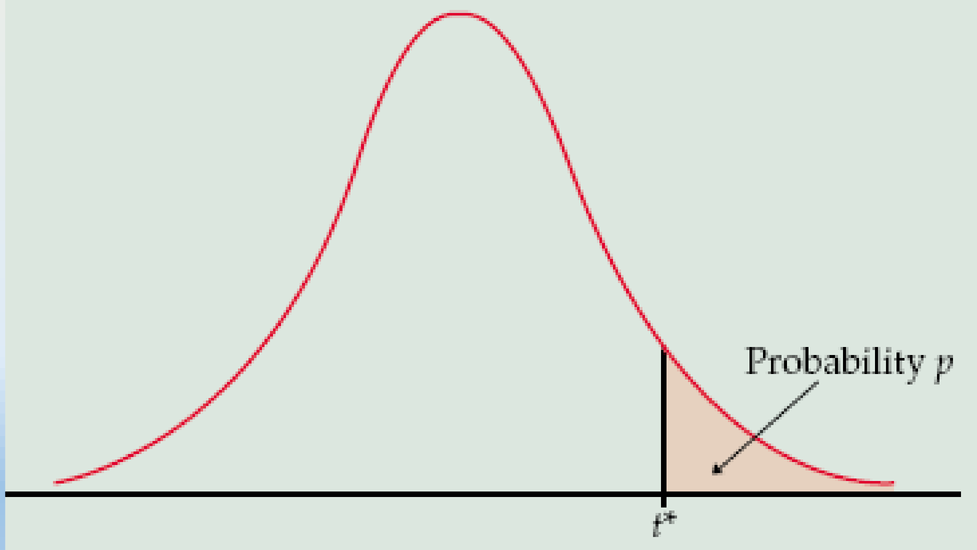
$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646

Smaller samples

T-tables

Table of the ***t*-distribution**, $P(t > t^*) = p$



df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646

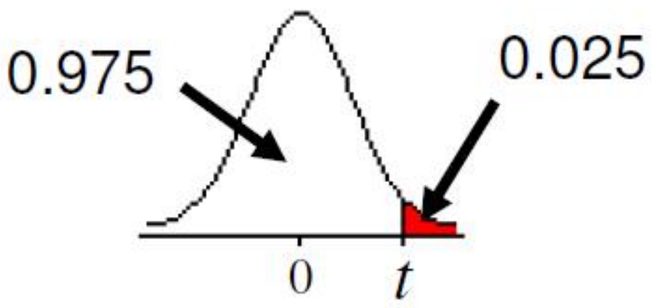
Smaller samples

How to use T-tables

Example, given a sample size of **n =10**, and I need value of **t** where the , $P(t > t) = 0.025$

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$
1	3.078	6.314	12.706	31.821
2	1.886	2.920	4.303	6.965
3	1.638	2.353	3.182	4.541
4	1.533	2.132	2.776	3.747
5	1.476	2.015	2.571	3.365
6	1.440	1.943	2.447	3.143
7	1.415	1.895	2.365	2.998
8	1.397	1.860	2.306	2.896
9	1.383	1.833	2.262	2.821
10	1.372	1.812	2.228	2.764
11	1.363	1.796	2.201	2.718
12	1.356	1.782	2.179	2.681
13	1.350	1.771	2.160	2.650
14	1.345	1.761	2.145	2.624
15	1.341	1.753	2.131	2.602

$$P(t_9 > t) = 0.025$$



From t-table,

$t = 2.262$

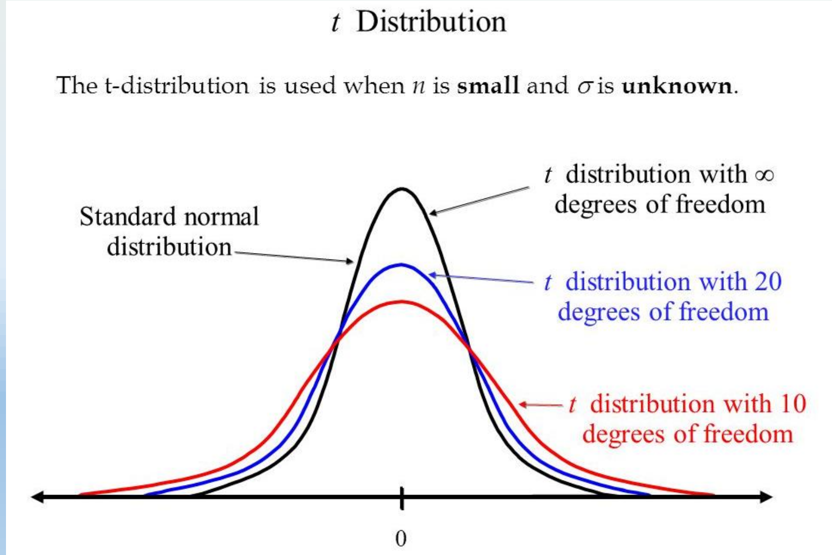
T-distribution is used in inferential statistics

Smaller samples

Z vs T

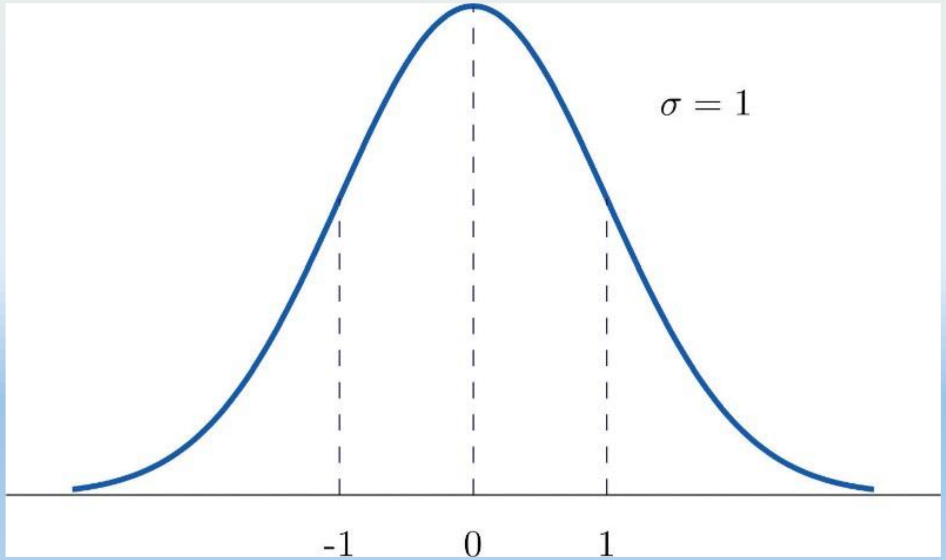
T-Distribution

- The general rule of thumb for when to use a **t score** is when your sample:
  - Has a sample **size below 30**,
  - Has an unknown population standard deviation.
- Affected by **degrees of freedom** because it is affected by sample size.



Z-Distribution

- Has a **sample size >=30**,
- Is not affected by degrees of freedom because you have one distribution for any sample above 30.



Samples & Sampling distributions

SUMMARY

✓Statistical studies – Process

- Formulate question
- Collect Data
- Answer question (by inference from data)
- Types of studies

✓Probability Sampling Methods:

- Random Sampling
- Systematic Sampling
- Cluster Sampling
- Stratified Random Sampling

✓Probability models for sampling distributions

- Central Limit Theorem
- For large samples, sampling distribution of the sample mean will be a normal distribution
- For small samples, approximate using the T-distribution **(Won't be examinable)**